

Creating AMI From EC2 Instance

Lab Details

1. This lab walks you through the steps to create an AMI from an Amazon EC2 instance. You will practice using Amazon Machine Images to launch Amazon EC2 instances and will create an AMI of your EC2 Instance.
2. Duration: **30 minutes**
3. AWS Region: **US East (N. Virginia) us-east-1**

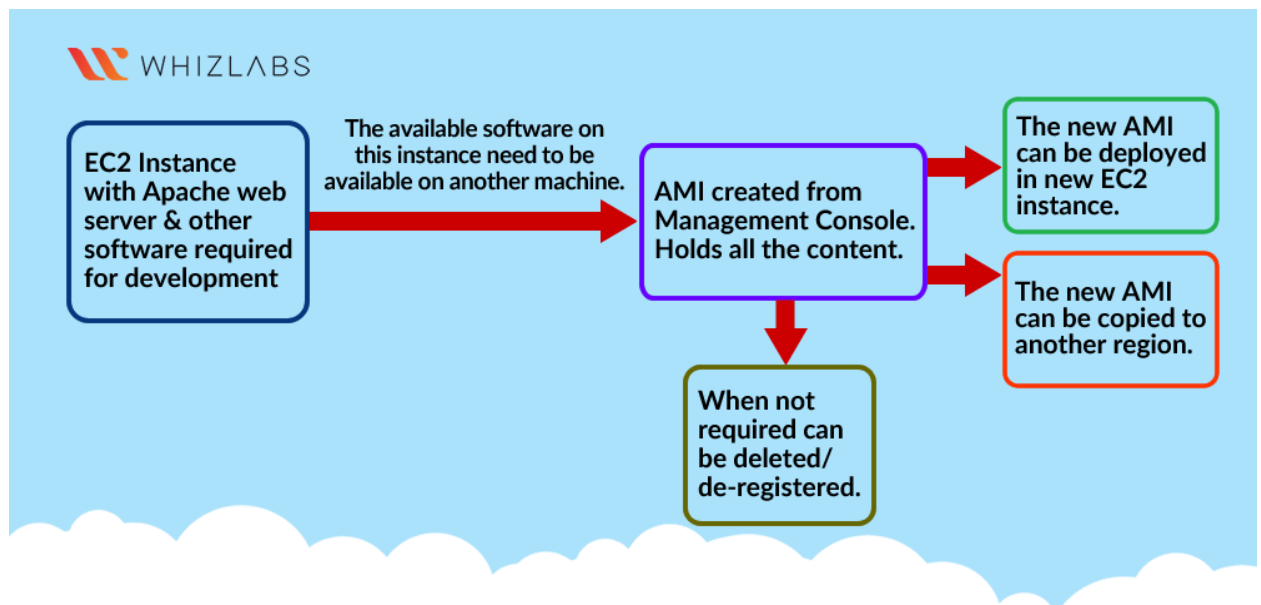
Introduction

What is AMI?

- AMI stands for Amazon Machine Image.
- It's a master image for the creation of virtual servers i.e., EC2 instances in the AWS environment.
- They are like templates that are configured with an operating system and other software, which determine the user's operating environment.
- AMIs are categorized according to region, operating system, system architecture (32 or 64 bit), launch permissions and whether they are backed by **Amazon EBS** or by the **Instance Store**.
- AMI includes a template for the root volume required for an instance; typical example might contain an operating system, an application server and applications.
- When you launch an instance, the root device volume contains the image used to boot the instance.
- In the initial stages, all AMIs were backed by the Amazon EC2 Instance Store. This means the root device for an instance launched from the AMI is an Instance Store volume created from a template stored in Amazon S3.
- Any data on the instance store volumes persists as-long-as the instance is running i.e., the data gets deleted once the instance is **terminated**.
- Instance store backed instances do not support the **Stop** action.
- If using an instance store is required, Amazon recommends distributing the data across multiple Availability Zones.
- After the introduction of **Amazon EBS**, Amazon introduced AMIs that are backed by Amazon EBS i.e., the root device for an instance launched from the AMI is an Amazon EBS volume created from an EBS Snapshot.

- Amazon recommends using **EBS backed AMIs**, because they launch faster and use persistent storage.
- Amazon EBS backed instances can be stopped and later restarted without affecting data stored in the attached volumes.
- Permissions are controlled to constrain AMIs for instance launches to the appropriate AWS accounts.
- A block device mapping ensures that the correct volumes are attached to the launched instance.
- Users have the facility of selecting AMI provided by AWS, the user community, or through the AWS Marketplace.
- Users can also create their own AMIs and share them within the same region or across regions.

Architecture Diagram



Task Details

1. Log into the AWS Management Console.
2. Create an EC2 Instance.
3. Create a new AMI using the EC2 Instance.
4. Checking the new EC2 Instance created with AMI.
5. Launching the EC2 instance created with AMI.
6. Validation of the lab.

Lab Steps

Task 1: Launching EC2 Instance

1. Make sure you are in the **US East (N. Virginia) us-east-1** Region.
2. Navigate to **Services** menu at the top, then click on **EC2** in the  **Compute** section.
3. Navigate to **Instances** on the left panel. Click on Instances and click **Launch instances**.
4. **Choose an Amazon Machine Image (AMI)** : Search for **Amazon Linux 2 AMI** in the search box and click on the **select** button.

Step 1: Choose an Amazon Machine Image (AMI)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace, or you can select one of your own AMIs.



5. **Choose an Instance Type** : Select **t2.micro** **Free tier eligible** and click on

Next: Configure Instance Details

6. Configure Instance Details:
 - Number of instances: Enter **1**
 - Auto-assign Public IP: Select **Enable**
 - Click on **Advanced Details**.
 - Under the **User data** section, enter the following script (which creates an HTML page served by an Apache httpd web server).

Copy

```
#!/bin/bash

sudo su

yum update -y

yum install httpd -y

echo "<html><h1> Welcome to Whizlabs </h1><html>" >>
/var/www/html/index.html

systemctl start httpd
```

```
systemctl enable httpd
```

Next: Add Storage

7. Click on

Next: Add Tags

8. **Add Storage**: No changes needed, click on

9. **Add Tags** : For identification of your instances, you can add a tag with key pair combination

- Key: Enter **Name**
- Value: Enter **MyEC2Server**

Next: Configure Security Group

- Click on

10. **Configure Security Group** : Select **Create a new security group**,

- Security group name: Enter **MyEC2SecurityGroup**
- Description : Enter **My EC2 Security Group**
- To add **SSH**:

◦ Choose Type: SSH

◦ Source: Anywhere

- For **HTTP**:

Add Rule

◦ Click on

◦ Choose Type: **HTTP**

◦ Source: Anywhere

- Click on

Review and Launch

Launch

11. **Review and Launch** : Review all your settings and click on

12. **Key Pair**: Select **Proceed without a keypair** and Check the checkbox to acknowledge.

Launch Instances

13. Click on

14. **Launch Status**: Your instance is now launching, Navigate to **Instances** page from the left menu and wait until the status of the EC2 Instance changes to running.

☐	Name ▾	Instance ID	Instance state ▾	Instance type ▾
☐	MyEC2Server	i-0a34a178ba5e84b1b	✓ Running 🔍	t2.micro

15. Note down the sample IPv4 Public IP Address of the EC2 instance.

Details
Security
Networking
Storage
Status checks
Monitoring
Tags

▼ Instance summary Info

Instance ID
i-0a34a178ba5e84b1b (MyEC2Server)

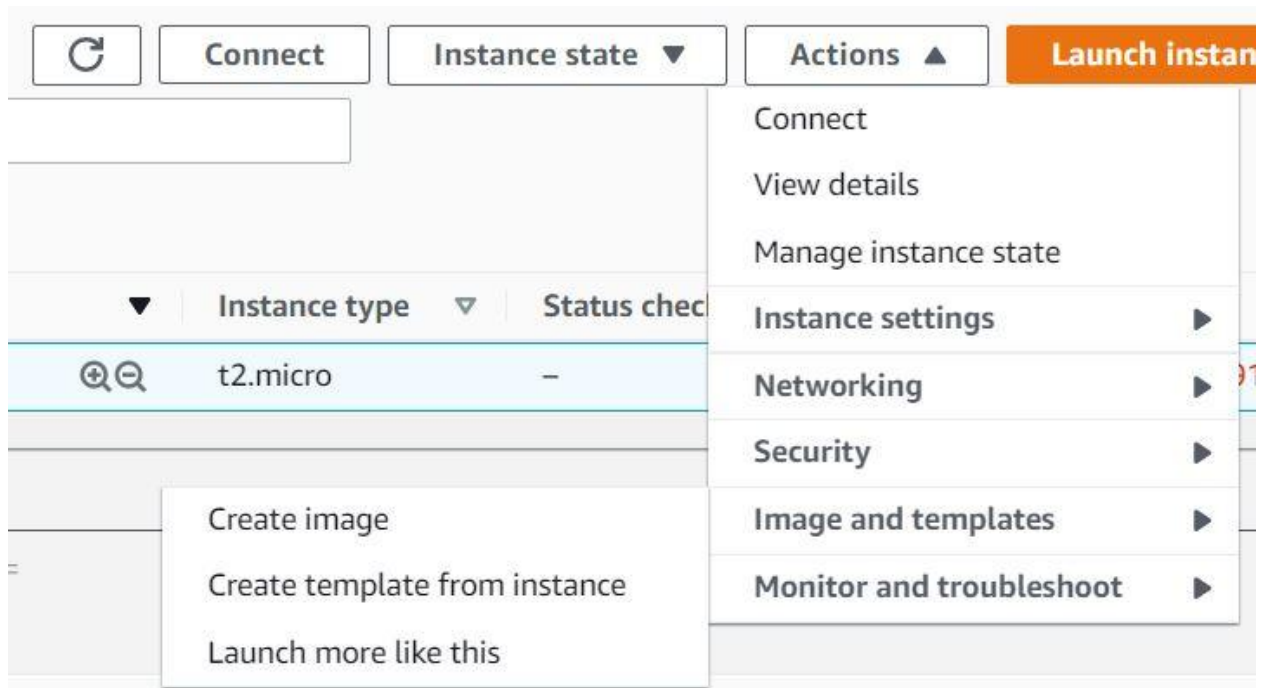
Public IPv4 address
3.80.149.180 | [open address](#)

16. Enter **http://your IP address**



Task 2: Creating an AMI from the EC2 Instance

1. Select the **MyEC2Server**. Click on **Actions**.
2. Under **Image and templates**, click on **Create Image**.



3. In the pop up window, enter the following details:
 - Image Name : Enter **MyEC2Image**
 - Image Description : Enter **My EC2 Image**
 - Leave other details as default.
 - Click on **Create image**.

Task 3: Check the newly created Image

1. Navigate to **AMI's** under **Images** on the left menu.

	Nam	AMI Name	AMI ID	Source	Owner	Visibility	Status
		MyEC2Image	ami-02516d21ccf560b...	757712384777/MyEC2Im...	757712384777	Private	pending

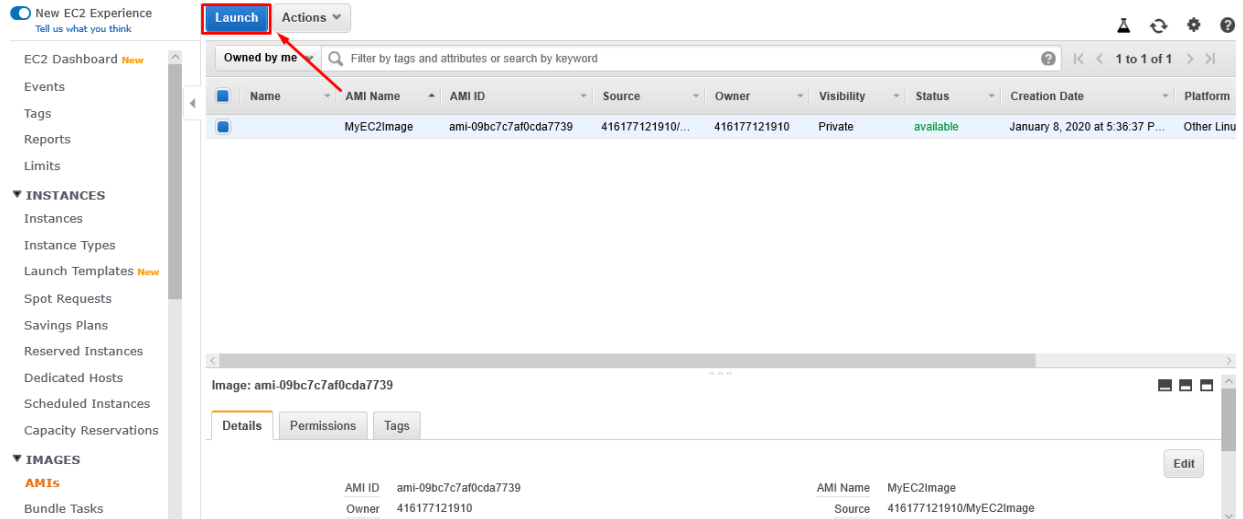
2. You can see that the image is getting generated and status is **pending**.
3. Once the process is completed, the status will change to **available**.

	Nam	AMI Name	AMI ID	Source	Owner	Visibility	Status
		MyEC2Image	ami-02516d21ccf560b...	757712384777/MyEC2Im...	757712384777	Private	available

4. Now we can use this Image AMI to create brand new instances.

Task 4: Launching an EC2 Instance with the Created AMI and Testing the AMI

1. Select the AMI and click on **Launch**.



2. Choose an Instance Type : Select **t2.micro** and click on

Next: Configure Instance Details

3. Configure Instance Details:

- Auto-assign Public IP: Select **Enable**
- No need to change anything.

Next: Add Storage

4. Click on

Next: Add Tags

5. Add Storage: No changes needed, click on

6. Add Tags : For identification of your instances, you can add a tag via key pair combination

- Key : Enter **Name**
- Value: Enter **MyEC2Server2**

Next: Configure Security Group

- Click on

7. Configure Security Group : Click on **Select the existing security group**,

select **MyEC2SecurityGroup** and click on

Review and Launch

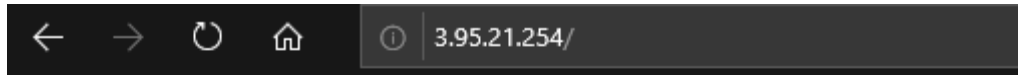
8. Review and Launch : Review all your settings and click on

Launch

9. Key Pair: Select **Proceed without a keypair** and Check the checkbox to acknowledge.

Launch Instances

10. Click on .
11. Navigate to the instances menu and copy the IPv4 Public IP address of the created EC2 instance.
12. Enter the IP Address in the Browser.



Welcome to Whizlabs